

Smart Lock Base Units

Full Product Range

Australian Compliance-Ready Mechanical Chassis — Supplier Specification

智能锁机械底座产品规格书（澳大利亚合规版） — 供应商技术规格

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Purpose / 目的	This edition adds diagrams and a bilingual glossary to reduce ambiguity for manufacturing partners. / 本版本增加了图示和中英文对照术语表，以减少制造合作伙伴的理解歧义。

1. Executive Summary / 概述

EBIZCO Australia is looking for a manufacturing partner to produce **five mechanical lock base units**. EBIZCO supplies the electronics and software. The supplier supplies the mechanical lock body and factory-standard electronics (host controller, motor, keypad, fingerprint sensor). All five lock types must physically and electrically accept EBIZCO's own communication module.

EBIZCO 澳大利亚正在寻找制造合作伙伴生产五种机械锁底座。EBIZCO 提供电子模块和软件；供应商提供机械锁体和厂标电子件（主控芯片、电机、键盘、指纹模块）。五种锁型都必须能够安装 EBIZCO 自己的通信模块。

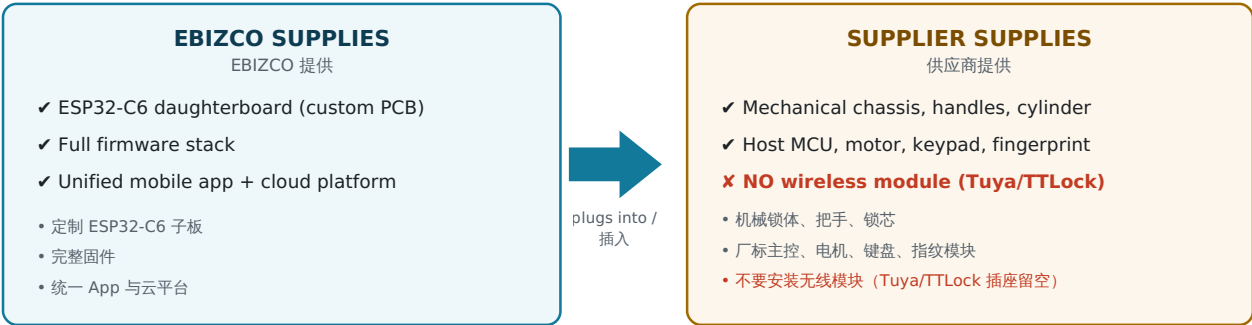


Figure 1 — Division of responsibility between EBIZCO and the supplier / 图1: EBIZCO 与供应商职责划分

CRITICAL / 关键要求

The wireless-module socket (for Tuya/TTLock-type modules) must be left **completely empty** on every electronic lock. EBIZCO's own daughterboard plugs into this same socket/position instead.

所有电子锁型的无线模块插座（Tuya/TTLock 类型）必须保持**完全空置**。EBIZCO 的子板将插入同一位置。

2. Product Catalogue Required / 所需产品目录

Five lock types make up the full commercial range. Types 1-4 are electronic; the padlock (Type 5) may be mechanical-only or electronic.

全系列共五种锁型。1-4 号为电子锁；5 号挂锁可以是纯机械款或电子款。



Figure 2 — The five required lock types / 图2: 所需的五种锁型

No.	Lock Type / 锁型	Application / 应用场景
1	Timber Door Lock / 木门锁	Standard residential/commercial wooden doors / 标准住宅或商用木门
2	Glass Door Lock (Push & Slide) / 玻璃门锁 (推拉式)	Glass doors, shopfronts, office partitions / 玻璃门、店面、办公室隔断
3	Rim Lock / 明装锁	Surface-mounted; retrofit on thin doors / 表面安装, 适用于薄门改造
4	Deadbolt / 插芯保险锁	Secondary security lock, standalone or with latch / 辅助安全锁, 可独立或配合门锁使用
5	Padlock / 挂锁	Gates, storage units, lockers / 大门、仓储柜、储物柜

3. Scope of Supply per Lock Type / 每种锁型的供货范围

✓ Required ⦿ Optional ✗ Not required / must be empty

Component / 部件	Timber	Glass	Rim	Deadbolt	Padlock
Lock body / chassis / 锁体	✓	✓	✓	✓	✓
Handle / lever (in & out) / 内外把手	✓	✓	✓	✗	✗
Mortise / cylinder mechanism / 锁芯机构	✓	✓	✓	✓	✓
Motorised drive / 电机驱动	✓	✓	✓	✓	⦿
Capacitive keypad / 电容键盘	✓	✓	✓	✓	⦿
Fingerprint sensor / 指纹模块	✓	✓	✓	✓	⦿
Host MCU (factory) / 厂标主控	✓	✓	✓	✓	⦿
Wireless module (Tuya/TTLock) / 无线模块	✗	✗	✗	✗	✗

CRITICAL / 关键要求

All electronic lock types must expose a **plaintext UART (TX/RX)** for EBIZCO's daughterboard. The wireless-module socket must remain **unpopulated** — do not solder a Tuya/TTLock module in place.

所有电子锁型必须留出**明文 UART (TX/RX)** 接口供 EBIZCO 子板使用。无线模块插座**不得贴片**——请勿焊接 Tuya/TTLock 模块。

4. Material Requirements by Tier / 分级材质要求

RESIDENTIAL TIER 住宅级 — 独立屋、联排别墅	COMMERCIAL TIER 商用级 — 公寓、酒店、写字楼、物业
Lock body / chassis: Aluminium or Zinc alloy	Lock body / chassis: Grade 304 Stainless Steel
Latch tongue / bolt: Zinc or Aluminium	Latch tongue / bolt: Grade 304 Stainless Steel
Internal drive gears: Zinc or Aluminium	Internal drive gears: Grade 304 Stainless Steel
锁体/门舌/内部齿轮：铝合金或锌合金	锁体/门舌/内部齿轮：304 不锈钢

Figure 3 — Material tiers / 图3：材质分级

5. Mandatory Compliance Requirements / 强制合规要求

5.1 Rule 1 — Instant Emergency Exit (NCC D2.21 / AS1905.1) / 紧急逃生要求

Applies to: Timber, Glass, Rim Locks (where an internal handle is present).

适用于：木门锁、玻璃门锁、明装锁（凡有内侧把手的锁型）。



Figure 4 — Emergency exit: required one-step release vs. banned two-step release / 图4：紧急逃生——必须的一步式与禁止的两步式对比

Requirement / 要求	Specification / 规格
Inside handle operation / 内侧把手操作	Single downward push / 单次向下推压
Deadbolt override / 保险栓解除	Automatic — no separate snib / 自动解除，无需单独旋钮
Latch & deadbolt / 门闩与保险栓	Retract simultaneously / 同时缩回
BANNED / 禁止	Separate thumb-turn or any intermediate step / 单独转动旋钮或任何中间步骤

5.2 Rule 2 — Disabled-Friendly Handles (AS1428.1) / 无障碍把手要求

Applies to: Timber, Glass, Rim Locks (where handles are present).

适用于：木门锁、玻璃门锁、明装锁（凡有把手的锁型）。

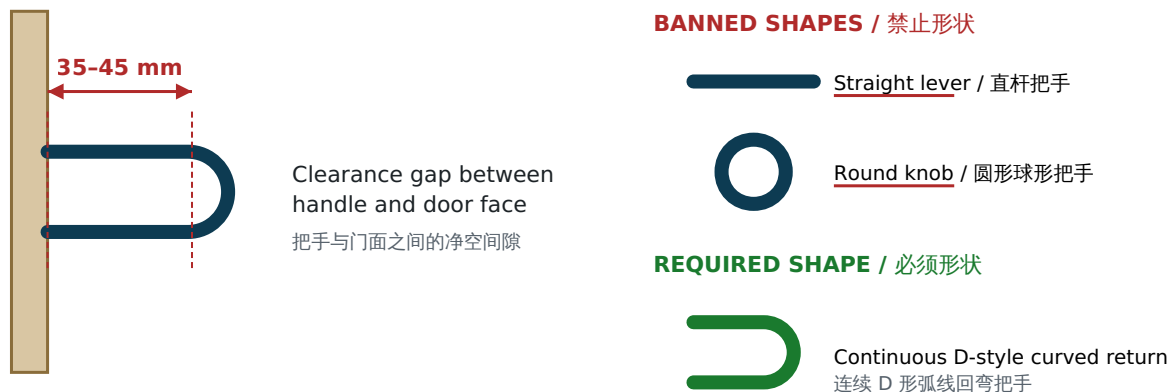


Figure 5 — Handle clearance dimension and required D-style shape / 图5：把手净空尺寸与必须的D形把手样式

Requirement / 要求	Specification / 规格
Handle design / 把手设计	Continuous D-style curved return / 连续D形弧线回弯
Clearance gap / 净空间隙	35-45 mm between handle and door face / 把手与门面之间35-45毫米
BANNED / 禁止	Straight levers, round knobs / 直杆把手、圆形球形把手

5.3 Rule 3 — Fire Survival (AS1530.4) — Commercial Tier Only / 耐火要求 (仅商用级)

Applies to: All lock types installed on fire-rated doors.

适用于：所有安装在防火门上的锁型。

Requirement / 要求	Specification / 规格
Latch tongue / bolt / 门舌	Grade 304 or 316 Stainless Steel / 304 或 316 不锈钢
Internal drive gears / 内部齿轮	Grade 304 or 316 Stainless Steel / 304 或 316 不锈钢
BANNED / 禁止	Zinc alloy, aluminium alloy internals / 内部零件使用锌合金或铝合金
Fire test documentation / 检测报告	Supplier to provide AS1530.4 test report / 供应商须提供 AS1530.4 检测报告

6. Technical Requirements — Electronic Locks / 电子锁技术要求

6.1 Electronic Interface (Common Across All Types) / 通用电子接口

Requirement / 要求	Specification / 规格
Host MCU / 主控芯片	Factory-standard microcontroller / 厂标标准微控制器
Communication / 通信方式	Plaintext UART (TX/RX) / 明文 UART (TX/RX)
Protocol / 协议	Standard Tuya MCU Serial Protocol (0x55 0xAA headers, 8-bit checksum) / 标准涂鸦 MCU 串口协议
Wireless module / 无线模块	Socket left EMPTY / 插座留空
Connector / 连接器	4-pin JST-PH compatible (VBAT_IN, GND, TX, RX) / 兼容4针JST-PH连接器

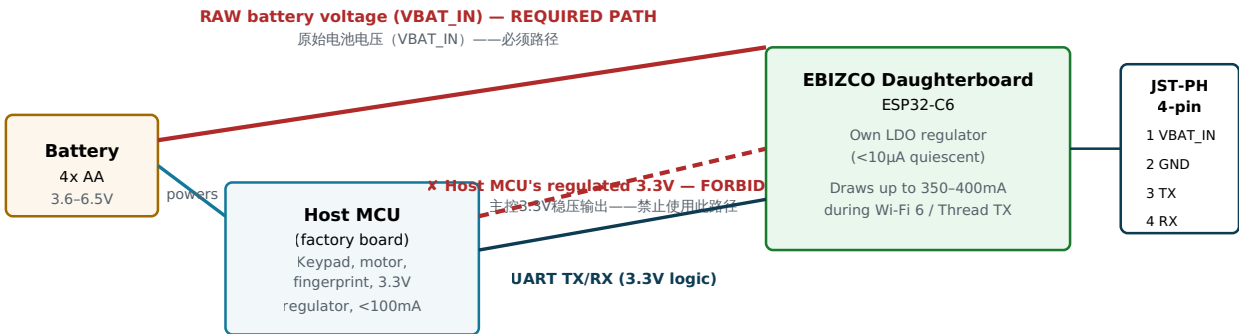


Figure 6 — Power & signal architecture: the daughterboard draws power directly from the raw battery, never from the host MCU's 3.3V rail / 图6：电源与信号架构——子板必须直接从电池取电，绝不可使用主控的3.3V输出

WHY THIS MATTERS / 为何如此重要

EBIZCO's ESP32-C6 daughterboard can draw **350-400mA** during Wi-Fi/Thread transmission bursts. A typical host MCU's 3.3V rail is current-limited to under 100mA and will brown out if it also has to power the daughterboard. The daughterboard has its own ultra-low-quiescent-current LDO regulator built in — it only needs the **raw, unregulated battery voltage** on VBAT_IN.

EBIZCO 的 ESP32-C6 子板在 Wi-Fi/Thread 发射时电流峰值可达 350-400mA。而主控芯片的 3.3V 输出通常电流限制在 100mA 以内，若为子板供电会导致主控掉电重启。子板内置了自己的超低静态电流 LDO 稳压器，只需要 VBAT_IN 引脚提供**未经稳压的原始电池电压**即可。

6.2 Power Supply Interface — Pinout Detail / 电源接口引脚详情

Pin	Signal	Voltage	Requirement / 要求
1	VBAT_IN	3.6V–6.5V DC	Raw battery voltage. Trace width ≥ 25 mil to support 500mA peak. Bypass capacitor $\geq 10\mu\text{F}$ optional. / 原始电池电压，走线宽度 ≥ 25 mil，支持500mA峰值电流
2	GND	0V	Common ground, trace width ≥ 25 mil / 公共地，走线宽度 ≥ 25 mil
3	TX	3.3V	Module $\rightarrow$ MCU (full duplex). Connect to host MCU UART RX pin. / 模块 $\rightarrow$ 主控，接主控UART RX引脚
4	RX	3.3V	MCU $\rightarrow$ Module (full duplex). Connect to host MCU UART TX pin. / 主控 $\rightarrow$ 模块，接主控UART TX引脚

Optional keep-alive handshake (per Tuya standard) / 可选唤醒握手信号：

Signal	Direction	GPIO	Description / 说明
SRDY	MCU $\rightarrow$ Module	GPIO3	MCU wakes module (active low) / 主控唤醒模块（低电平有效）
MRDY	Module $\rightarrow$ MCU	GPIO20	Module wakes MCU / has data (active low) / 模块唤醒主控 / 有数据（低电平有效）

6.3 Power Consumption Guidelines / 功耗设计参考

Component / 部件	Peak Current (Typical) / 峰值电流（典型值）
Host MCU + Keypad + Fingerprint / 主控+键盘+指纹	30–50mA @ 3.3V
Motor (unlocking burst) / 电机（解锁瞬间）	200–300mA @ 6V (sourced separately / 独立供电)
Daughterboard ESP32-C6 (Wi-Fi TX) / 子板（Wi-Fi 发射）	350–400mA @ 3.6V (after LDO / 经LDO稳压后)

6.4 Mechanical Interface per Type / 各锁型机械接口

Lock Type / 锁型	Key Mechanical Specifications / 关键机械规格
Timber Door Lock / 木门锁	Tubular latch: 60mm or 70mm backset; door thickness 35–45mm / 管式门闩，60或70毫米后距，门厚35–45毫米

Glass Door Lock (Push & Slide) / 玻璃门锁	Push & slide mechanism; compatible with 8-12mm glass / 推拉机构, 适配8-12毫米玻璃
Rim Lock / 明装锁	Surface-mounted; standard Australian backset / 表面安装, 标准澳洲后距
Deadbolt / 插芯保险锁	Single or double-cylinder; 60mm or 70mm backset / 单锁芯或双锁芯, 60或70毫米后距
Padlock / 挂锁	Shackle diameter/clearance per Australian standard; electronic optional / 锁梁直径及净空按澳洲标准, 可选电子款

7. Pricing & Volumes / 价格与数量

Parameter / 参数	Value / 数值
Initial order (combined) / 首批订单 (合计)	1,000 units, mix of types / 1000件, 多种锁型混合
Target price — Residential / Aluminium / 目标价格——住宅级/铝合金	USD \$12-20 per unit, stripped, no wireless module / 每件12-20美元, 不含无线模块
Target price — Commercial / 304 Stainless / 目标价格——商用级/304不锈钢	USD \$35-50 per unit, stripped, no wireless module / 每件35-50美元, 不含无线模块
Incoterms / 贸易条款	FOB or EXW (please confirm) / FOB或EXW (请确认)

Prices to be quoted per lock type. Volume mix to be confirmed upon sample validation. / 请按锁型分别报价。数量混合比例将在样品验证后确认。

8. Sampling & Validation / 样品与验证

Item / 项目	Requirement / 要求
Sample quantity / 样品数量	2-3 units per lock type / 每种锁型2-3件
Sample delivery / 样品运输	Airmail / Express / 空运或快递
Validation / 验证方式	Full compliance testing in Australia / 在澳大利亚进行全面合规检测
Supplier documentation / 供应商文件	Datasheet, photos, test reports for each type / 每种锁型提供数据表、照片、检测报告

9. Supplier Response Checklist / 供应商回复清单

Please confirm the following for **each lock type** you can supply:

请针对**每种可供货锁型**确认以下事项：

1. Which lock types can you supply? (Timber / Glass / Rim / Deadbolt / Padlock) — 可供货的锁型
2. Model name / chassis name per type — 每种锁型的型号／锁体名称
3. Photos: exterior, internal cavity, mechanism — 外观、内部结构、机构照片
4. Inside handle video: latch + deadbolt retraction (for door locks) — 内侧把手视频：门闩与保险栓缩回演示（门锁类）
5. Side-profile photo with ruler showing handle clearance gap — 带尺子的侧面照片，展示把手净空间隙
6. Material type per type (Aluminium / Zinc / Stainless) — 每种锁型的材质（铝／锌／不锈钢）
7. Stripped price (no Tuya module) — per type and for 1,000-unit mix — 不含Tuya模块的裸价（按锁型及1000件混合总价）
8. AS1530.4 fire test report availability — 是否具备AS1530.4耐火检测报告
9. Lead time for samples — 样品交期
10. Lead time for bulk order (1,000 total) — 批量订单交期（合计1000件）

10. Bilingual Quick-Reference Glossary / 中英文术语速查表

Key technical terms used throughout this document, for quick reference by engineering and production teams. / 本文件中出现的关键技术术语速查，供工程与生产团队参考。

English	中文	Note / 说明
Host MCU	厂标主控芯片	Supplier's own microcontroller, factory-standard
Daughterboard	子板 / 附加板	EBIZCO's ESP32-C6 communication module
Plaintext UART	明文串口通信	Not encrypted; standard TX/RX serial lines
Wireless module socket left empty	无线模块插座留空	Do not populate with Tuya/TTLock module
Raw battery voltage (VBAT_IN)	原始电池电压	Unregulated voltage straight from the battery pack
Regulated 3.3V rail	3.3V稳压输出	Host MCU's own power output — must NOT power the daughterboard

Brownout	电压跌落 / 掉电重启	Voltage sag causing the MCU to reset or misbehave
LDO regulator	低压差线性稳压器	Low-dropout regulator, built into the daughterboard
Backset	后距	Distance from door edge to the centre of the lock cylinder
Thumb-turn / snib	拇指旋钮 / 保险旋钮	A separate manual lock/unlock knob — banned on emergency exit paths
D-style curved return handle	D形弧线回弯把手	Required handle shape for disabled-access compliance
Stripped price	裸价 / 不含模块价格	Price excluding the wireless communication module
FOB / EXW	离岸价 / 工厂交货价	Incoterms for shipping responsibility

11. Contact / 联系方式

Name / 姓名	Vince Phan
Title / 职位	Systems Architect Director
Company / 公司	EBIZCO Australia Pty Ltd
Email / 邮箱	[email]
Phone / 电话	[phone]

This document outlines EBIZCO's minimum requirements across the full product range. Full PRS and engineering specifications will be shared upon supplier confirmation.

本文件概述了 EBIZCO 对全系列产品的最低要求。供应商确认合作后，将提供完整的PRS及工程规格文件。